

ARF project proposals 2024-2025 (MIARFID)

1 Multimodal Classification of IDH1 Mutation with Missing Modality Robustness

Team members

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Brief description

This project focuses on the multimodal classification of the IDH1 genetic mutation in Glioblastoma patients using the UPenn-GBM dataset. In real-world clinical settings, acquiring all imaging modalities is often impossible due to time constraints, patient movement, or missing equipment. Therefore, our primary objective is to develop a robust multimodal neural network capable of handling missing inputs, known as "Modality Dropout".

The dataset contains four 3D NIfTI MRI volumes (T1, T1-Gd, T2, T2-FLAIR) and tabular clinical data per patient. We will train a model using all available MRI sequences fused with tabular patient data to establish a strong classification baseline. During testing, we will simulate missing modalities by dynamically dropping specific MRI sequences (e.g., replacing T1-Gd and T2-FLAIR with zeros) and evaluating the model's ability to maintain predictive accuracy using only the remaining scans and tabular features.

The project workflow will involve preprocessing 3D medical images, implementing a multimodal fusion architecture, integrating a modality dropout layer during training, and evaluating performance under varying degrees of missing inputs.

List of objectives

1. To collect and prepare the UPenn-GBM dataset (NIfTI volumes and clinical CSV)
2. To define experimental design: preprocess 3D MRI volumes and tabular data, defining training, dev and test sets

3. To select and design a multimodal neural network architecture supporting modality dropout
4. To implement the selected model and run experiments establishing a baseline with all modalities
5. To run inference experiments simulating missing MRI sequences
6. To analyse results and evaluate the model's robustness against missing data

Subset of objectives for project review (29/04/26): 1-4